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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 22	
TITLE : JDBC PreparedStatement, and ResultSet	

Problem Statement

Demonstration and exploration of JDBC classes and interfaces (PreparedStatement, and ResultSet).

Learning Objectives

To implement PreparedStatement and ResultSet for secure query execution and data retrieval.

Programme

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;

public class JDBCPreparedStatement {

   public static void main(String[] args) {

      String url = "jdbc:mysql://localhost:3306/college";
      String user = "root";
      String password = "Husam123";

      try {
         Connection con = DriverManager.getConnection(
            url, user, password
         );

         System.out.println("Database connected successfully.");

         String query = "SELECT id, name, marks FROM student WHERE marks >= ?";

         PreparedStatement ps = con.prepareStatement(query);
```

```
        ps.setDouble(1, 50);

        ResultSet rs = ps.executeQuery();

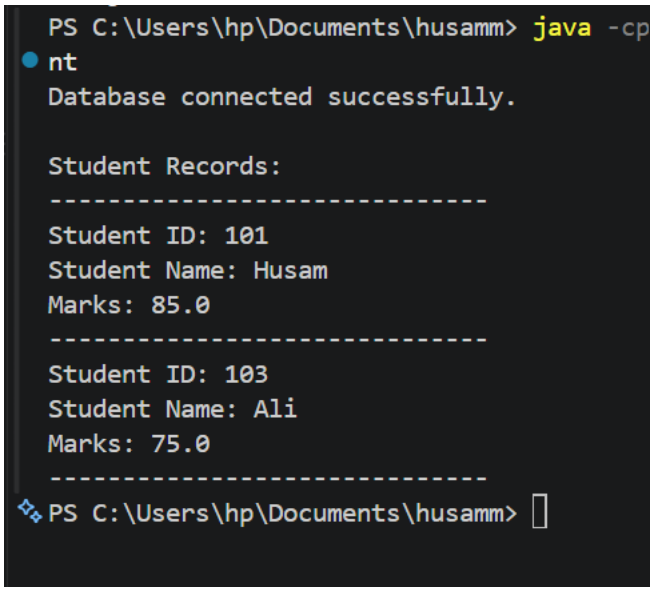
        System.out.println("\nStudent Records:");
        System.out.println("-----------------------------");

        while (rs.next()) {
            System.out.println("Student ID: " + rs.getInt("id"));
            System.out.println("Student Name: " + rs.getString("name"));
            System.out.println("Marks: " + rs.getDouble("marks"));
            System.out.println("-----------------------------");
        }

        rs.close();
        ps.close();
        con.close();

    } catch (SQLException e) {
        System.out.println("Error: " + e.getMessage());
    }
  }
}
```

Output



Exercise 1. Create a login application using PreparedStatement to check username and password.

Code

```java
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.Scanner;

public class LoginApplication {

    Run | Debug
    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/college";
        String user = "root";
        String password = "Husam123";

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Username: ");
        String username = sc.nextLine();

        System.out.print("Enter Password: ");
        String loginPassword = sc.nextLine();

        String query = "SELECT * FROM users WHERE username = ? AND password = ?";

        try {
            Connection con = DriverManager.getConnection(
                url, user, password
            );

            PreparedStatement ps = con.prepareStatement(query);

            ps.setString(1, username);
            ps.setString(2, loginPassword);

            ResultSet rs = ps.executeQuery();

            if (rs.next()) {
                System.out.println("Login Successful!");
                System.out.println("Welcome, " + username);
            } else {
                System.out.println("Invalid Username or Password.");
            }

            rs.close();
            ps.close();
            con.close();

        } catch (SQLException e) {
            System.out.println("Database Error: " + e.getMessage());
        }
```

Output

```
avaC:\Users\hp\Documents\husamm>
PS C:\Users\hp\Documents\husamm> java -cp
Enter Username: husam
Enter Password: 12345
Login Successful!
Welcome, husam
PS C:\Users\hp\Documents\husamm>
```

2. Create a Java application using PreparedStatement to authenticate hospital staff login (doctor/nurse/login ID and password). Display appropriate access messages based on authentication result.

Code

HospitalStaffLogin.java > HospitalStaffLogin > main(String[])

```java

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.Scanner;

public class HospitalStaffLogin {

    Run | Debug
    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/college";
        String dbUser = "root";
        String dbPassword = "Husam123";

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Login ID: ");
        String loginId = sc.nextLine();

        System.out.print("Enter Password: ");
        String password = sc.nextLine();

        String query = "SELECT role FROM hospital_staff " +
                       "WHERE login_id = ? AND password = ?";

        try {
            Connection con = DriverManager.getConnection(
                url, dbUser, dbPassword
            );

            PreparedStatement ps = con.prepareStatement(query);

            ps.setString(1, loginId);
            ps.setString(2, password);

            ResultSet rs = ps.executeQuery();

            if (rs.next()) {
                String role = rs.getString("role");

                if (role.equalsIgnoreCase("Doctor")) {
                    System.out.println("Authentication Successful!");
                    System.out.println("Welcome Doctor. Access Granted.");
                } else if (role.equalsIgnoreCase("Nurse")) {
                    System.out.println("Authentication Successful!");
```

```
32
33              PreparedStatement ps = con.prepareStatement(query);
34
35              ps.setString(1, loginId);
36              ps.setString(2, password);
37
38              ResultSet rs = ps.executeQuery();
39
40              if (rs.next()) {
41                  String role = rs.getString("role");
42
43                  if (role.equalsIgnoreCase("Doctor")) {
44                      System.out.println("Authentication Successful!");
45                      System.out.println("Welcome Doctor. Access Granted.");
46                  } else if (role.equalsIgnoreCase("Nurse")) {
47                      System.out.println("Authentication Successful!");
48                      System.out.println("Welcome Nurse. Access Granted.");
49                  } else {
50                      System.out.println("Access Denied: Invalid staff role.");
51                  }
52
53              } else {
54                  System.out.println("Authentication Failed!");
55                  System.out.println("Invalid Login ID or Password.");
56                  System.out.println("Access Denied.");
57              }
58
59              rs.close();
60              ps.close();
61              con.close();
62
63          } catch (SQLException e) {
64              System.out.println("Database Error: " + e.getMessage());
65          }
66
67          sc.close();
68      }
69  }
70
```

Output

```
PS C:\Users\hp\Documents\husamm> java -c
Enter Login ID: DOC101
Enter Password: doc123
Authentication Successful!
Welcome Doctor. Access Granted.
PS C:\Users\hp\Documents\husamm>
```

Conclusion

The program demonstrates the use of JDBC PreparedStatement and ResultSet to execute SQL queries securely and retrieve data from a MySQL database. It successfully displays the required records and helps prevent SQL injection through parameterized queries.